Lattice Before Ring: A Two-Level Silicon Photonic Demultiplexer, Benchmarked and Tested for Fabrication Tolerance

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edX public username: wzwqoo · SiEPIC openEBL, 220 nm SOI, four cells of 605 × 410 µm

Abstract: This is the short version of the design report, written for peer review. Four layouts were submitted, each with a separate job: All four pass the EBeam DRC and SiEPIC verification with zero errors. Every device was simulated in Lumerical, and its expected spectrum is given here.

- **Chip 1 (EBeam_wzwqoo_design):** a two-level demultiplexer. An MZI lattice tree splits four coarse channels 10 nm apart, and microrings followed by contra-directional couplers (CDCs) pick four fine 200 GHz channels. A second copy of the ring bank uses apodised CDCs.
- **Chip 2 (EBeam_wzwqoo_benchmark):** the same coarse job done three other ways (an AWG, an echelle grating, a cascade of CDCs), plus a Fabry–Pérot cavity.
- **Chip 3 (EBeam_wzwqoo_FabChallenges):** fabrication checks. CDC-A drawn too narrow, as designed and too wide, with all four ports on grating couplers; the width-test MZIs at nominal width; two ring pairs, one of identical rings to expose manufacturing error and one of different radii that passes a single line; and the grating-coupler loopback.
- **Chip 4 (EBeam_wzwqoo_FabChallenges2):** the width tests. The MZIs drawn too narrow and too wide, and the F2 ring at 500 nm and at a matched width, each drawn too narrow, as designed and too wide.

Keywords: wavelength-division multiplexing, MZI lattice filter, microring resonator, contra-directional coupler, arrayed waveguide grating, Vernier ring pair, fabrication tolerance

1. Design Concept

1.1. The problem

Optical links carry many wavelengths on one fibre, and the receiver must separate them. Two kinds of filter are needed:

- a **coarse** filter that splits wide bands (here 10 nm apart) with a **flat** top, so everything inside a band passes equally;
- a **fine** filter that picks single channels 200 GHz (1.6 nm) apart with a **narrow** line.

No single filter does both well. An AWG, for example, would need at least 50 channel slots, a 370 µm star-coupler radius and about 160 arms to cover both grids, far more than the 605 µm cell holds.

1.2. The two-level architecture (chip 1)

- **Coarse level: a 1×4 MZI lattice tree (D1).** Each stage is a chain of unbalanced MZIs. Its coupler ratios shape a flat-topped passband 8.57 nm wide, wider than the 4.78 nm the fine channels occupy.
- **Fine level: four add-drop microrings (D2).** Each ring ($R \approx 7.5$ µm) drops one 200 GHz channel with a 0.2 nm wide line. Its FSR (12.1 nm) is larger than the 10 nm coarse spacing, so in the full system the lattice in front removes the ring's other orders.
- **Order selector on the test chip: CDCs.** On chip 1 the rings are measured without the lattice in front. A CDC after each ring passes a single band (it has no FSR), so the ring's other orders are removed on the chip as well.
- **Apodised copy (D3).** D2's uniform CDCs keep side lobes above -20 dB for tens of nm on both sides of their band, right where the rings' other orders sit. D3 repeats D2 with both CDCs apodised, to measure how much of that apodisation removes on the same channels.

1.3. The benchmarks (chip 2) and the fabrication tests (chips 3 and 4)

Chip 2 does the coarse job of D1 with three other filters, on the same four channels C1–C4, so the four can be compared on one test station.

- **AWG-C (arrayed waveguide grating).** An input slab spreads the light over 27 waveguide arms, each 6.982 µm longer than the last. The growing delay tilts the wavefront leaving the arms, and the output slab focuses each wavelength onto a different output guide. It is the standard dense demultiplexer in telecom. The PDK has no AWG cell, so it was drawn from the design rules of Smit and van Dam [7].
- **EDG-C (echelle diffraction grating).** The AWG folded into one slab: instead of arms, a row of 28 etched facets reflects the light back and focuses each wavelength onto its own output [10]. A 3-pair Bragg mirror behind each facet raises its reflection [11]. It is also drawn, as the PDK has no cell.
- **CDC cascade.** Three PDK contra-directional couplers on one bus, each dropping one coarse channel and passing the rest to the next [12]. They are apodised (the grating strength tapers off at both ends) to lower the side lobes that would otherwise leak into the neighbouring channels.
- **FP cavity.** Two PDK Bragg mirrors with a half-period spacer between them pass a single narrow line inside the mirrors' stop band [6]. It asks whether a single-line filter can separate two neighbouring fine channels, which a CDC band (9 nm wide) cannot.

Chips 3 and 4 repeat the most width-sensitive devices with every drawn width, and so every gap, offset by d: −15, 0 and +8 nm for the rings and CDC-A, −15, 0 and +15 nm for the MZI delays. Width error is the largest fabrication variable, and on chip 1 the F2 ring line sits only 0.48 nm inside CDC-A's band.

Chip 3 also carries two ring pairs, each two add-drop rings in series. In TwinRings the rings are identical, so every line passes at full height only if the fab prints both alike, the assumption behind D2's ring bank. In Vernier the radii differ, so the two combs meet only once in the laser sweep and the pair passes a single line, a second single-line filter beside the FP cavity.

1.4. Wavelength plan

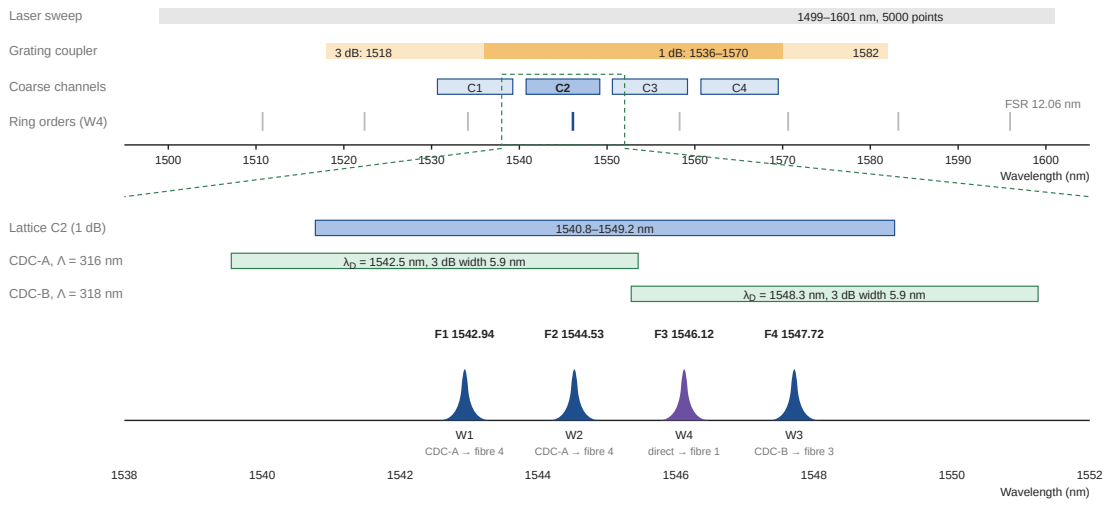


Fig. 1. Wavelength plan against the test window. Top: laser range, grating-coupler band, the coarse channels C1–C4, and the other orders of one ring. Bottom: zoom on C2 with the CDC bands and the fine channels F1–F4.

- **Coarse channels C1–C4: 1535, 1545, 1555, 1565 nm.** The spacing is 10 nm, not the 20 nm CWDM grid, because the grating couplers lose too much at the band edges: a 20 nm grid would lose 9.96 dB on its outer channels, against 2.56 dB here.
- **Fine channels F1–F4: 1542.94, 1544.53, 1546.12, 1547.72 nm.** They are on the ITU 200 GHz grid, inside C2, the flattest part of the grating-coupler response.
- **Fine linewidth: 0.2 nm.** It gives about 10 laser points per line with the 20.4 pm test-laser step, and −24 dB between neighbouring channels.

1.5. What each device tests

Table 1. The 26 devices, grouped by type

Type	Label	Device	Chip	Question it answers
Lattice	LatticeD1	1×4 MZI lattice tree	1	Are the coarse bands flat and isolated?
Rings	RingCDCD2	4 rings + CDC-A + CDC-B	1	Do the rings land on F1–F4, and does the CDC remove their other orders?
Rings	RingCDCApo	D2 again with both CDCs apodised, W1 left out (D3)	1	How much does apodisation lower the CDC side lobes next to the dropped lines?
Rings	RingM15, Ring0, RingP8	the F2 ring (500 nm), width −15 / 0 / +8 nm	4	Does the ring line move with the CDC band, or away from it?
Rings	RingMatchM15, RingMatch0, RingMatchP8	F2 ring redrawn at 479 nm width, −15 / 0 / +8 nm	4	Does a ring of matched width stay inside the CDC band?
Rings	TwinRings	two identical rings in series, R = 7.529 µm, 500 nm	3	Do two neighbouring rings print alike?
Rings	Vernier	two rings in series, R = 7.529 and 8.232 µm, both 500 nm	3	Can two rings of different FSR pass a single line?
CDCs	CDCAM15, CDCA0, CDCAP8	CDC-A, width −15 / 0 / +8 nm; drop, through and add each on a grating coupler	3	How far does the CDC band move, in reflection and in transmission?
CDCs	CDCCascade	three apodised CDCs on one bus	2	Benchmark against the lattice
MZIs	MZIwire0, MZIwireM15, MZIwireP15	MZI, delay in 500 nm wire, width 0 / −15 / +15 nm	3, 4	How far does a width error move the fringes?
MZIs	MZIwide0, MZIwideM15, MZIwideP15	the same MZI, delay in a 3 µm guide	3, 4	Does a wide delay guide remove that drift?
MZIs	MZIwide0T40	MZIwide0 with the PDK's 40 µm tapers instead of 50 µm	3	Does the shorter taper cost anything?
Gratings	AWGC	1×4 arrayed waveguide grating	2	Benchmark against the lattice
Gratings	EDGC	1×4 echelle grating, Bragg-mirror facets	2	Benchmark against the lattice
Gratings	FPCavity	two Bragg mirrors + half-period spacer	2	Can a single narrow line separate two neighbouring fine channels?
Reference	GCLoopback	two GCs joined by a waveguide	3	Grating-coupler response, to normalise every spectrum

The label column is the opt_in_TE_1550_device_wzwqoo_<label> text on each input grating coupler.

2. Design Equations and Parameters

Each subsection gives the formulas that size a device and the values drawn in the layout. The formulas are from Chrostowski and Hochberg [1] unless another source is given; values marked "solved" come from Lumerical (MODE FDE, EME, varFDTD or 3D FDTD).

2.1. Strip waveguide

Every device is built from the 500 × 220 nm strip. Its group index sets every spectral scale (FSR, linewidth), and any index change δn, from width, thickness or temperature, moves every filter by the drift law:

$$n_g = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}, \qquad \delta\lambda = \frac{\lambda\, \delta n}{n_g} \tag{1}$$

Table 2. Strip waveguide, solved in MODE FDE

Quantity	Value
$n_{\text{eff}}(\lambda)$, λ in µm	$2.44198 - 1.12195\,(\lambda - 1.55) - 0.03475\,(\lambda - 1.55)^2$
n_g at 1550 nm	4.181 (PDK model 4.191 [8], −0.24 %)
Width sensitivity	$\partial n_{\text{eff}}/\partial W$ = 1.574 × 10^{-3} nm^{-1} → a filter moves 0.581 nm per nm of width, by (1)
Thickness sensitivity	$\partial n_{\text{eff}}/\partial H$ = 3.705 × 10^{-3} nm^{-1} → 1.37 nm per nm of Si thickness

2.2. MZI lattice tree (D1, chip 1)

Each stage is a 2-channel half-band filter [2]: four delay sections between five directional couplers. The delay ΔL sets the stage FSR, which must be twice the channel spacing it separates (20 nm for stage 1, 40 nm for stage 2):

$$\text{FSR} = \frac{\lambda^2}{n_g\,\Delta L} \tag{2}$$

The integer order m then puts the passband peak exactly on the channel λ_A [9]:

$$m = \text{round}\left(\frac{n_{\text{eff}}\,\Delta L}{\lambda_A}\right) \;\Longrightarrow\; \Delta L_{\text{drawn}} = \frac{m\,\lambda_A}{n_{\text{eff}}(\lambda_A)} \tag{3}$$

Each coupler of power cross-coupling K is the PDK ebeam_dc_te1550 (two 500 nm guides, gap 200 nm) with a straight length L_c:

$$K = \sin^2\!\left(\frac{\pi(L_c + L_0)}{2L_x}\right) \tag{4}$$

where L_x is the cross-over length and L_0 the coupling already done in the S-bends.

Table 3. Lattice parameters

Parameter	Value
Delays by (2), (3): stage 1 / 2A / 2B	FSR 20 / 40 / 40 nm; λ_A = 1535 / 1535 / 1545 nm; m = 46 / 23 / 23; ΔL = 28.741 / 14.370 / 14.530 µm
π phase offset in the third section	$L_\pi = \lambda_A/(2n_{\text{eff}})$ = 0.312–0.316 µm
Coupler ratios $K_0 \ldots K_4$	0.500, 0.0730, 0.0658, 0.500, 0.2154
L_x, L_0 at 1550 nm (solved, 3D FDTD of the PDK cell)	38.29 µm, 2.04 µm
Straight lengths L_c, by (4)	17.106, 4.628, 4.284, 17.106, 9.724 µm

Why these coupler ratios. The published set of Dwivedi et al. [3] (0.50, 0.13, 0.12, 0.50, 0.25) leaks −14.1 dB of the other channel within ±3 nm of each channel centre. The set above was found by minimising that worst leak and reaches −25.9 dB, with an 8.57 nm flat top.

2.3. Microrings (D2 and D3, chip 1)

Four add-drop rings on one bus, each built from two PDK half-rings (ebeam_dc_halfring_straight, point-coupled). The radius sets the FSR and the channel; the gap sets the coupling κ^2 and so the linewidth [4]:

$$\text{FSR} = \frac{\lambda^2}{n_g\,2\pi R}, \qquad \text{FWHM} \approx \frac{\text{FSR}\,\kappa^2}{\pi} \tag{5}$$

A ring's Lorentzian line leaks into a neighbouring channel $\Delta\lambda$ away by

$$\text{leak} = \frac{1}{1 + (2\Delta\lambda/\text{FWHM})^2} = \frac{1}{1 + (2 \times 1.593/0.2)^2} = -24.1\ \text{dB} \tag{6}$$

All four rings: width 500 nm, order m = 75, gap 110 nm for κ^2 = 0.049 per coupler (FWHM 0.2 nm with 3 dB/cm loss assumed; drop loss −0.28 dB). The PCell rounds each radius to the 1 nm grid, so the rings are drawn at 7.516, 7.529, 7.542 and 7.555 µm, which moves each line by at most 0.06 nm. One 200 GHz channel step needs 13.25 nm more

Table 4. Ring parameters

Ring	Channel	Radius	Bent n_{eff} (solved)	FSR	Drop routed to
W1	F1, 1542.94 nm	7.5157 µm	2.45053	12.056 nm	CDC-A → fibre 4
W2	F2, 1544.53 nm	7.5289 µm	2.44874	12.059 nm	CDC-A → fibre 4
W4	F3, 1546.12 nm	7.5422 µm	2.44695	12.062 nm	fibre 1, direct
W3	F4, 1547.72 nm	7.5555 µm	2.44516	12.066 nm	CDC-B → fibre 3

radius. The rings are tuned by radius rather than width, because on the 1 nm layout grid a width step moves a channel by 36 % of a spacing and a radius step by only 7.5 %. D3 has W4, W3 and W2 only, so its CDC-A carries F2 alone: in D2, W1 and W2 share CDC-A, so W1 adds no second comparison, and leaving it out saves the 20 µm the column needs.

2.4. Contra-directional couplers CDC-A and CDC-B (chip 1)

The PDK contra_directional_coupler couples a forward mode in one guide to a backward mode in the other at one band, dropped back towards the input side [5]. Its centre and each guide's own (unwanted) Bragg reflection are

$$\lambda_D = (n_1 + n_2)\,\Lambda, \qquad \lambda_{B,i} = 2\,n_i\,\Lambda \tag{7}$$

and its peak drop grows with the grating length L as $\tanh^2(\kappa L)$.

Table 5. CDC parameters

Parameter	CDC-A	CDC-B
Guides, corrugation	560 / 440 nm wide, teeth 50 / 25 nm	same
Gap	116 nm (PDK default 100 nm puts facing teeth 62.5 nm apart, below the 70 nm minimum Si space)	same
Period Λ	316 nm	318 nm
Periods N, length, apodisation	D2: 300, 94.8 µm, none (uniform); D3: 250, about 80 µm, index 1.3	same
λ_D by (7), with the solved $n_1 + n_2$	1542.54 nm ($n_1 + n_2$ = 4.8815)	1548.28 nm
Self-Bragg λ_{B1} / λ_{B2} by (7)	1579.72 / 1508.70 nm	1585.88 / 1514.06 nm
Grating strength κ (solved, EME)	$\approx$ 0.040 µm^{-1} → κL = 3.8, peak drop 0.998	same grating, solved for CDC-A only
Channels carried	F1 and F2 (D3: F2 only)	F4

The apodised CDCs of D3. Both take apodisation index 1.3 and N = 250, the printable taper of chip 2's cascade (Section 2.8); the rings, gaps and routing are D2's. D3's CDC-B (Λ 318 nm) is the apodised C2 grating of the cascade. The apodised CDC-A (Λ 316 nm, whole-grating EME) drops at -0.13 dB with a 9.2 nm 3 dB band, centred 2.0 nm blue of the uniform grating. The period is kept at 316 nm on purpose, so that D3 differs from D2 in the apodisation alone; 1 nm more period would re-centre the band on F2.

2.5. FP cavity (chip 2)

Two PDK Bragg gratings (ebeam_bragg_te1550) act as mirrors. A half-period spacer between them opens one transmission line at the centre of their stop band [6]. The line narrows as the mirror reflection approaches 1:

$$\lambda_B = 2\bar{n}\,\Lambda, \qquad R = \tanh^2(\kappa L), \qquad L_s = \frac{\Lambda}{2} \tag{8}$$

The target line width is 0.2–0.32 nm: narrow enough to reject F1 by 20 dB, no narrower than the 0.2 nm ring line. 60-period mirrors (R = 0.46) give no usable line, and 200 periods narrow it to about 0.1 nm. 150 periods give 0.30 nm and reject F1 by 20.5 dB.

Table 6. FP cavity parameters

Parameter	Value
Mirror grating	Λ = 317 nm, corrugation ΔW = 50 nm, 500 nm guide (drawn alternating between 450 and 550 nm)
Grating strength κ (solved, EME of one 60-period mirror)	0.043 µm^{-1} (λ_B = 1544.3 nm, so $\bar{n}$ = 2.436 by (8))
Periods per mirror	150 (47.55 µm) $\rightarrow$ R = 0.94 by (8), line width 0.30 nm
Spacer L_s	158 nm (Λ/2 on the 1 nm grid)
Layout	one straight vertical piece, 95.3 µm long: a bend would stretch the spacer and open more lines

2.6. AWG-C (chip 2)

Three rules size an AWG [7]. The arm step ΔL sets the grating order and the FSR, the slab radius R_a sets the spacing of the channels at the output, and the arm count M must capture the beam that spreads from the input guide (half-width θ_0):

$$m = \frac{n_{\text{eff}}\,\Delta L}{\lambda_c}, \qquad \text{FSR} = \frac{\lambda_c^2}{n_g\,\Delta L} \tag{9}$$

$$R_a = \frac{d_o\, n_s\, d\, \lambda_c}{n_g\,\Delta L\,\Delta\lambda_{\text{ch}}} \tag{10}$$

$$M \geq \frac{3\,\theta_0\, R_a}{d} \tag{11}$$

Table 7. AWG-C parameters

Parameter	Value
Centre wavelength, channel spacing, channels	λ_c = 1550 nm, $\Delta\lambda_{\text{ch}}$ = 10 nm, 4 outputs
Order m, arm step ΔL, FSR, by (9)	11, 6.982 µm, 82.3 nm
Slab index n_s (solved, slab FDE)	2.8446
Arm pitch d, output pitch d_o	2.0 µm, 2.0 µm
Slab radius R_a, by (10)	60.44 µm
Beam half-width θ_0, arm count M, by (11)	0.29 rad, 27 arms
Arm and I/O tapers	0.5 $\rightarrow$ 1.8 µm over 10 µm, so 0.2 µm gaps where they meet the slab
Refocus (solved, varFDTD)	every I/O aperture moved 1.6 µm towards the grating, because the beam leaves the taper from a point 1.6 µm behind its mouth
Arm layout	27 nested-U arms, 161–342 µm long in steps of ΔL; 1.2 µm pitch (0.7 µm gaps) at the U tops and legs, coupling estimated at about $-$35 dB
Cell size	139 $\times$ 197 µm

2.7. EDG-C (chip 2)

The facets lie on a grating circle. The grating equation sets the facet pitch p for order m, and the circle radius R sets the channel spacing at the outputs [10]:

$$n_s\, p\,(\sin\theta_i + \sin\theta_d) = m\lambda, \qquad \text{FSR} = \frac{\lambda\, n_s}{m\, n_{gs}} \tag{12}$$

$$R \approx \frac{d_o\, n_s\, \lambda_c}{2\, n_{gs}\,\Delta\lambda_{\text{ch}}\,\sin\theta_B} \tag{13}$$

Table 8. EDG-C parameters

Parameter	Value
Slab index n_s, slab group index n_{gs} (solved)	2.846, 3.695
Order m, FSR, by (12)	14, 85.3 nm
Blaze angle θ_B, facet pitch p	45°, 5.39 µm
Grating-circle radius R, by (13); output pitch d_o	168.9 µm; 2.0 µm
Facets	28
Facet mirrors	3 quarter-wave pairs behind each facet: Si 136 nm, oxide 268 nm [11]
Cell size	108 × 191 µm

2.8. CDC cascade (chip 2)

Three PDK CDCs as in Section 2.4, sized by (7), one per channel, on one bus [12].

Table 9. CDC cascade parameters

Parameter	Value
Periods Λ (channel)	325 nm (C4), 321 nm (C3), 318 nm (C2); C1's CDC is left out, as only three output fibres exist
Periods N, length, gap	250, about 80 µm, 116 nm
Apodisation	PCell Gaussian index a = 1.3: every tooth at least 5 nm deep (an assumed print limit); largest side lobe −13.2 to −13.5 dB, against −4.1 to −4.7 dB uniform
Bus order	longest wavelength first, so no channel reaches a CDC whose self-Bragg reflection (about 37 nm above its drop) could send it back

If the shallowest teeth do not print. The apodised gratings keep every tooth at least 5 nm deep, an assumed print limit. As the worst case, the C2 grating (Λ 318 nm) was solved again by whole-grating EME with every tooth shallower than **10 nm** removed, as if those teeth did not print at all. The drop stays at **1546.4 nm** with **−0.21 dB** loss (−0.12 dB as drawn) and a 9.2 nm band (9.6 nm). The largest side lobe falls from **−13.2 to −16.2 dB**, and the worst leak within ±3 nm of the neighbouring channels changes by **less than 0.5 dB**. Only the drop at the neighbouring channel centres rises, from −32 to −24 dB. The cascade's isolation therefore does not depend on the shallowest teeth printing.

2.9. Width-offset copies and fabrication tests (chips 3 and 4)

Every width is offset by d (d > 0 is wider). A wider guide also narrows the gap beside it, so the gaps change by −d. Table 10 lists the values drawn in the GDS.

The MZIs. Each is built from PDK cells: an ebeam_y_1550 Y-branch as splitter and another as combiner. Each arm has four 500 nm bends (R = 5 µm) and two vertical delay legs. A leg is a 50 µm ebeam_taper_te1550 from 0.5 µm to the leg width, a straight, and a taper back; on the 3 µm legs the taper changes width by 2.5 µm over 50 µm, a 1:20 ratio. MZIwide0T40 keeps the PDK routing taper's 40 µm (1:16) for comparison. The arms are mirror images; the upper arm's straights are 7 µm and the lower arm's 2 µm, so ΔL = 2 × (7 − 2) = 10 µm, all of it in the leg width. Each arm's two legs are 14 µm apart: at 10 µm the taper's device-recognition regions overlapped.

The 3 µm legs stay single-mode. A 3 µm guide carries seven TE modes. An EME of the 50 µm taper shows that the higher modes together take only **−30.5 dB** (−27.8 dB at the PDK's 40 µm, almost all of it in TE2), and TE0 loses under **0.01 dB**. The odd modes are not excited, as the taper's symmetry requires, and any TE2 adds the same ripple to both arms. MZIwide0T40 measures the 40 µm version.

The two legs of an arm do not couple. They are 14 µm apart (an 11 µm gap). FDE supermodes of two 3 µm guides show their splitting falling tenfold per 0.23 µm of gap, so at 11 µm the coupling is far below anything measurable.

The rings. Two ebeam_dc_halfring_straight cells joined tip to tip, as on chip 1, with 10 µm ebeam_taper_te1550 tapers from the 500 nm routing to the ring width on all four bus ports. The largest step, 36 nm (RingMatchM15), is a 1:280 taper and adds no loss worth measuring. Input from the upper grating coupler, drop to the lower one; through and add end in terminators.

The CDCs. CDC-A's PCell (Λ 316 nm, N = 300, uniform) with both guide widths + d and the gap − d; its ports stay 500 nm. Drop, through and add each reach a grating coupler, so every CDC is read in reflection and in transmission on

Table 10. Chips 3 and 4, drawn values

Chip	Label	Copy of	d	Drawn widths and gap
3	CDCAM15 / CDCA0 / CDCAP8	CDC-A (560 / 440 nm, gap 116 nm)	−15 / 0 / +8 nm	545 / 425 nm, gap 131 nm / 560 / 440 nm, gap 116 nm / 568 / 448 nm, gap 108 nm
3	MZIwire0 / MZIwide0 / MZIwide0T40	MZI, delay in 500 nm wire / in 3 µm guide / the same with 40 µm tapers	0	delay legs 0.5 / 3.0 / 3.0 µm
3	TwinRings / Vernier	W2 ring, then an identical ring / a ring of R = 8.232 µm	0	500 nm, gap 110 nm
4	RingM15 / Ring0 / RingP8	F2 ring W2 (500 nm, gap 110 nm, R 7.529 µm)	−15 / 0 / +8 nm	485 nm, gap 125 nm / 500 nm, gap 110 nm / 508 nm, gap 102 nm
4	RingMatchM15 / RingMatch0 / RingMatchP8	F2 ring at W* = 479 nm, R = 7.639 µm	−15 / 0 / +8 nm	464 nm, gap 125 nm / 479 nm, gap 110 nm / 487 nm, gap 102 nm
4	MZIwireM15 / MZIwireP15	MZI, delay in 500 nm wire	−15 / +15 nm	delay legs 485 / 515 nm
4	MZIwideM15 / MZIwideP15	MZI, delay in 3 µm guide	−15 / +15 nm	delay legs 2.985 / 3.015 µm

the same device.

The ring pairs. The input runs straight into ring 1's bus; ring 1's drop feeds ring 2's bus 20 µm lower; ring 2's drop goes to the lower grating coupler. Ring 1 is the W2 ring (R = 7.529 µm, order 75). In TwinRings ring 2 is identical; in Vernier it has R = 8.232 µm, so that its order 82 also falls on F2.

The matched ring W*. The ring line moves 0.591 nm per nm of width, but CDC-A's band moves 0.667 nm/nm, because one CDC guide is only 440 nm wide and the gap shrinks as d grows. A 479 nm ring moves at 0.662 nm/nm, nearly with the CDC. Its radius grows from 7.529 to 7.639 µm to keep F2 at the same order.

Drawn widths, measured in the GDS. The 1 nm layout grid changes some odd widths (Section 4.1):

- **Rings: the offsets are exact.** The ring guide itself is drawn at the specified width: its area divided by πR gives 485.0, 479.0 and 487.0 nm for the odd widths. Only the straight bus is **1 nm narrower** (484, 478 and 486 nm), which widens the gap at the coupling point by 1 nm (126, 111 and 103 nm instead of 125, 110 and 102 nm). That lowers κ^2 by about 2 % ($\kappa^2 \propto e^{-g/53.5\ \mathrm{nm}}$ for the PDK half-ring) and leaves the ring's own index, and so its line, unchanged.
- **MZIs: the span is unchanged.** The delay legs are 484 / 514 nm and 2984 / 3014 nm instead of 485 / 515 and 2985 / 3015 nm, so the offsets are −16 and +14 nm. Both copies shift together, and their 30 nm span is unchanged; the d = 0 copies are exact.
- **CDC-A at −15 nm:** guides of 546 / 424 nm instead of 545 / 425 nm, with the mean gap at the designed 131 nm. The other two copies are exact (560 / 440 nm, gap 116 nm; 568 / 448 nm, gap 108 nm), and the closest facing teeth of the +8 nm copy are 71 nm apart, above the 70 nm minimum.

Why the ring and CDC copies stop at +8 nm. CDC-A's facing teeth are drawn 78.5 nm apart, and every nm of positive offset closes that space by one nm, so any offset beyond +8.5 nm breaks the 70 nm Si-space rule. Both device types use the same two offsets, so ring and CDC are always compared at the same d. The MZIs contain no narrow gaps and use the full ±15 nm.

3. Simulated Performance

Each component was solved in Lumerical (MODE FDE, EME, varFDTD or 3D FDTD), exported as S-parameters and assembled in INTERCONNECT with the PDK grating couplers. Fig. 4 shows every device's expected spectrum as the fibre measurement will see it, without and with its two grating couplers.

Main comparisons.

- **Coarse filters.** The lattice has the widest flat top (8.4–8.9 nm at 1 dB) and low loss. The AWG and the EDG isolate better but have rounded passbands only 2.8–4.0 nm wide. The CDC cascade loses least but leaks most (−16.5 dB).
- **Fine filters.** A plain ring drop leaks −24.1 dB into the neighbouring channel, as (6) predicts for a 0.2 nm line. A CDC after the ring removes the ring's other orders, not this neighbour leak.
- **Apodisation behind the rings (D3 against D2).** It lowers the leak of the other channels into CDC-A's fibre from about −30 to about −73 dB. It also moves CDC-A's band 2 nm blue, so F2, now near the upper edge, loses 1.7 dB

more: apodisation buys isolation with F2's margin.

- **Single-line filters.** The FP cavity's 0.30 nm line rejects F1 by 20.5 dB, so it can separate two neighbouring fine channels that a CDC band (9 nm wide) cannot. Vernier passes one 0.12 nm line by a different means (Section 5): it is narrower, but it magnifies any local difference between its two rings 11.7 times, so the FP cavity is the robust single-line filter and Vernier the sensitive gauge.

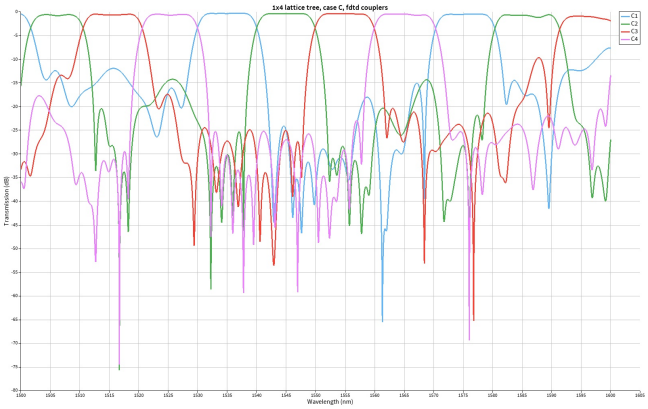


Fig. 2. Lattice tree D1, outputs C1–C4, couplers from 3D FDTD (INTERCONNECT).

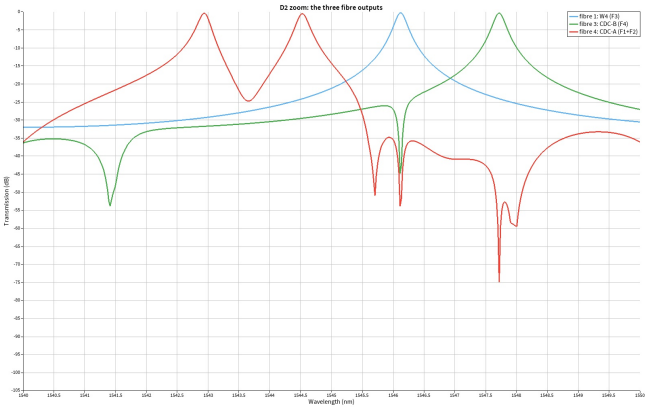


Fig. 3. Ring bank D2, 1540–1550 nm: W4 direct drop (fibre 1), CDC-B (fibre 3), CDC-A (fibre 4).

4. Layout and Verification

4.1. Layout

Manufacturability choices. Every component is a PDK cell or PCell, with only its parameters changed:

- **CDC gap 116 nm instead of the default 100 nm.** At 100 nm the facing teeth are only 62.5 nm apart, below the 70 nm minimum Si spacing; at 116 nm they are 78.5 nm apart.
- **CDC-A and CDC-B unapodised in D2.** The default apodisation tapers the end teeth to sub-nm depths, too shallow to print.
- **Cascade CDCs and D3's CDCs apodised at a = 1.3,** the strongest taper that keeps every tooth at least 5 nm deep (an assumed print limit, not a PDK rule; Section 2.8 checks the case where those teeth do not print).
- **Offsets stop at +8 nm** for the rings and CDCs on chips 3 and 4, where CDC-A's facing teeth reach the 70 nm limit.
- **MZI leg tapers at 1:20,** 50 µm for the 2.5 µm width step, longer than the PDK routing taper's 40 µm; MZIwide0T40 keeps 40 µm for comparison.
- **Ring gap 110 nm and coupler gap 200 nm,** both above the 70 nm minimum.
- **FP cavity drawn as one straight piece,** so the spacer keeps its 158 nm length (a bend would stretch it and open extra lines).
- **Odd widths are drawn 1 nm narrower on straight guides.** The layout grid is 1 nm, and the PDK cells place each waveguide edge at ± width/2 from its centre. For an odd width both edges of a straight guide land on a half-nanometre (487 nm → ±243.5 nm) and both round toward the centre, so a 487 nm straight is drawn as **486 nm**; even widths are exact. A ring's arc keeps its full width, because its vertices round independently. Only the width-offset copies have odd widths; Section 2.9 lists what this does to them.

Close waveguides. Silicon edges closer than 2 µm occur only where light is meant to couple (lattice couplers, ring gaps, CDC and Bragg teeth, the AWG and EDG tapers at their slabs) and in two places on chip 2:

- **AWG-C arms:** 1.2 µm pitch (0.7 µm gaps) at the U tops and legs, coupling between neighbours estimated at about −35 dB over the 60 µm U tops (not simulated).
- **Two routing pairs** run 1.7–1.8 µm apart for about 21–24 µm. At that gap two 500 nm strips exchange a negligible fraction of their power.

Device spacing. openEBL asks for at least 2 µm between devices and at least 45 µm between the grating-coupler black boxes of different devices; no DRC deck checks either, so both were measured on the layouts. The closest devices

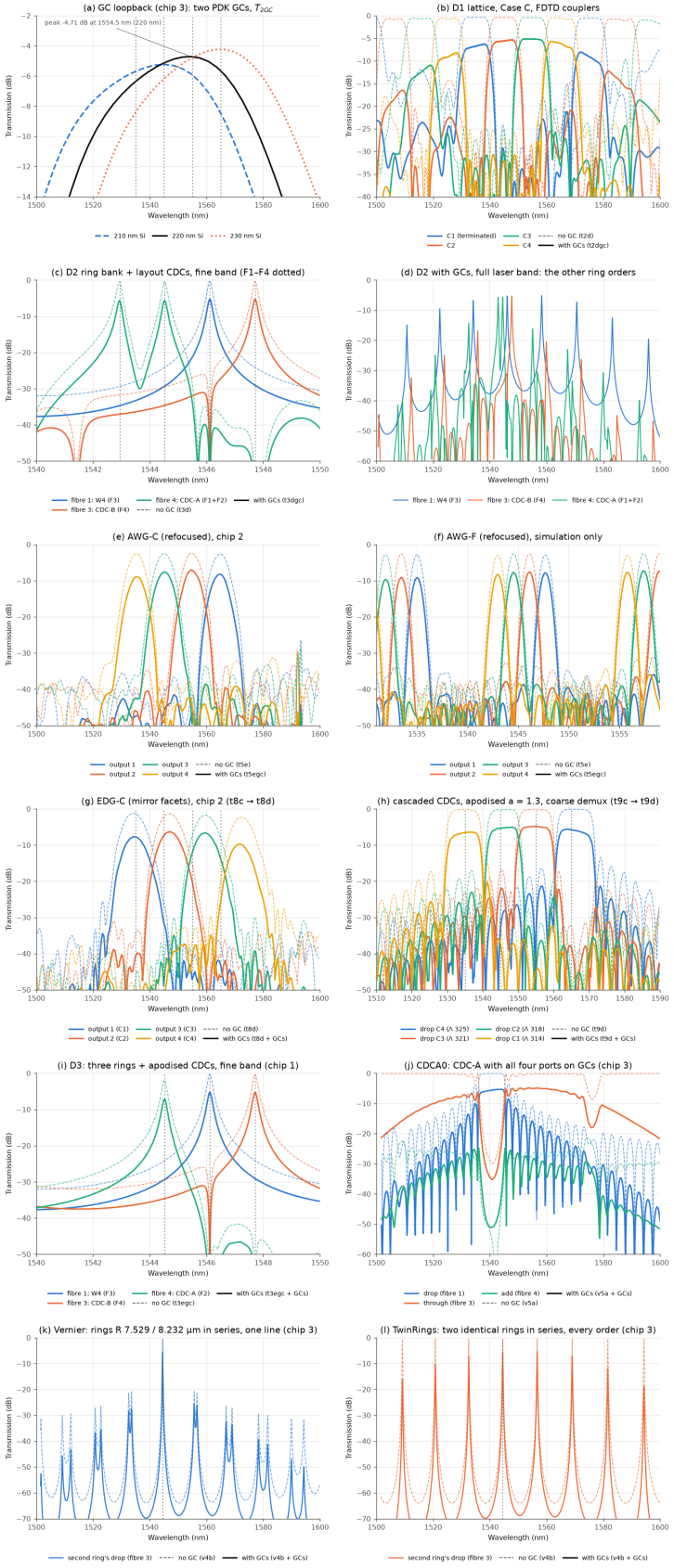


Fig. 4. What the fibre measurement will see: each device without (dashed) and with (solid) its two PDK grating couplers. (a) GC loopback at 210 / 220 / 230 nm Si. (b) Lattice D1. (c) Ring bank D2, fine band. (d) D2 over the full laser band. (e) AWG-C and (f) AWG-F (simulation only). (g) EDG-C. (h) CDC cascade. (i) D3, three rings with apodised CDCs, fine band. (j) CDCA0: drop, through and add. (k) Vernier: one line at F2. (l) TwinRings: every order.

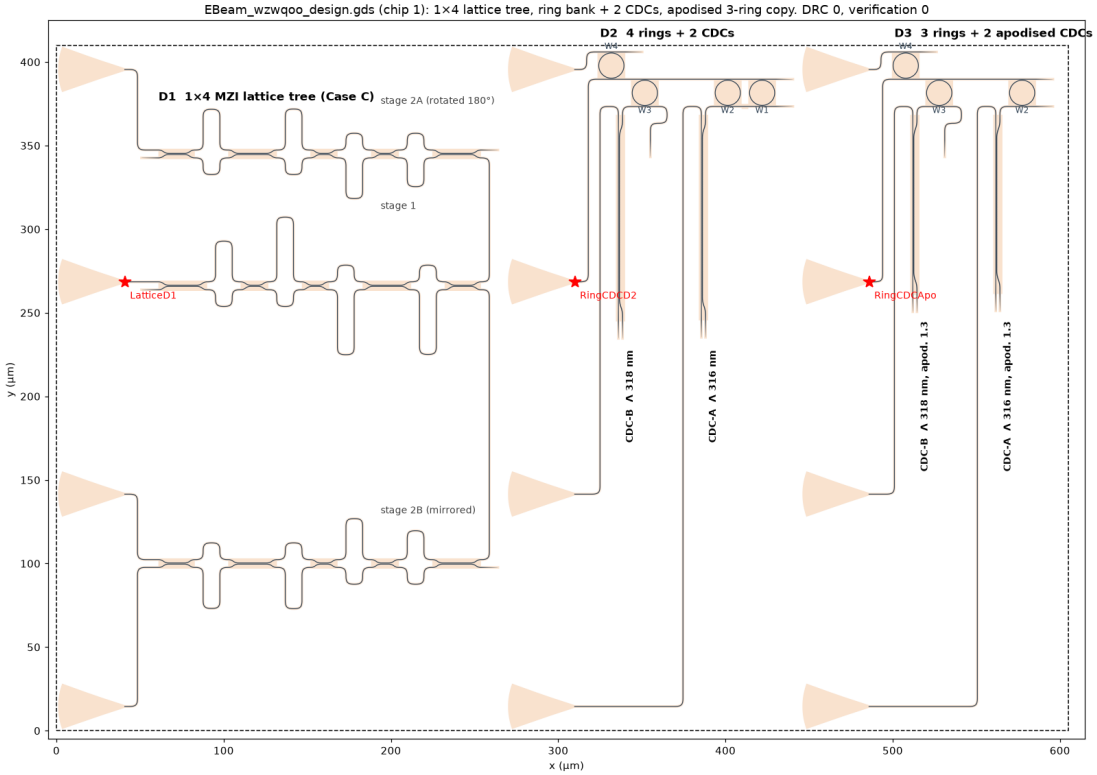


Fig. 5. Chip 1 (EBeam_wzwqoo_design). D1 lattice tree; D2 four rings and two CDCs; D3 three rings and two apodised CDCs.

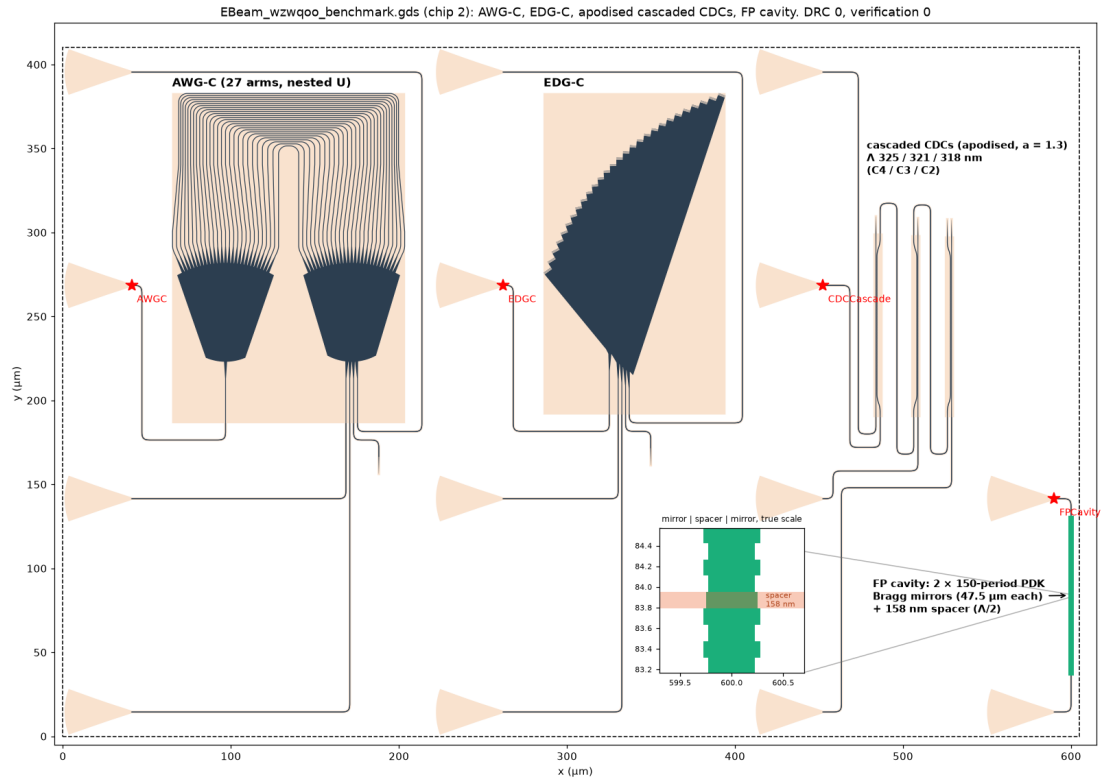


Fig. 6. Chip 2 (EBeam_wzwqoo_benchmark). AWG-C, EDG-C and three apodised cascaded CDCs, each with its own grating-coupler column, and the FP cavity at the right edge. Inset: the 158 nm spacer between the two Bragg mirrors at true scale.

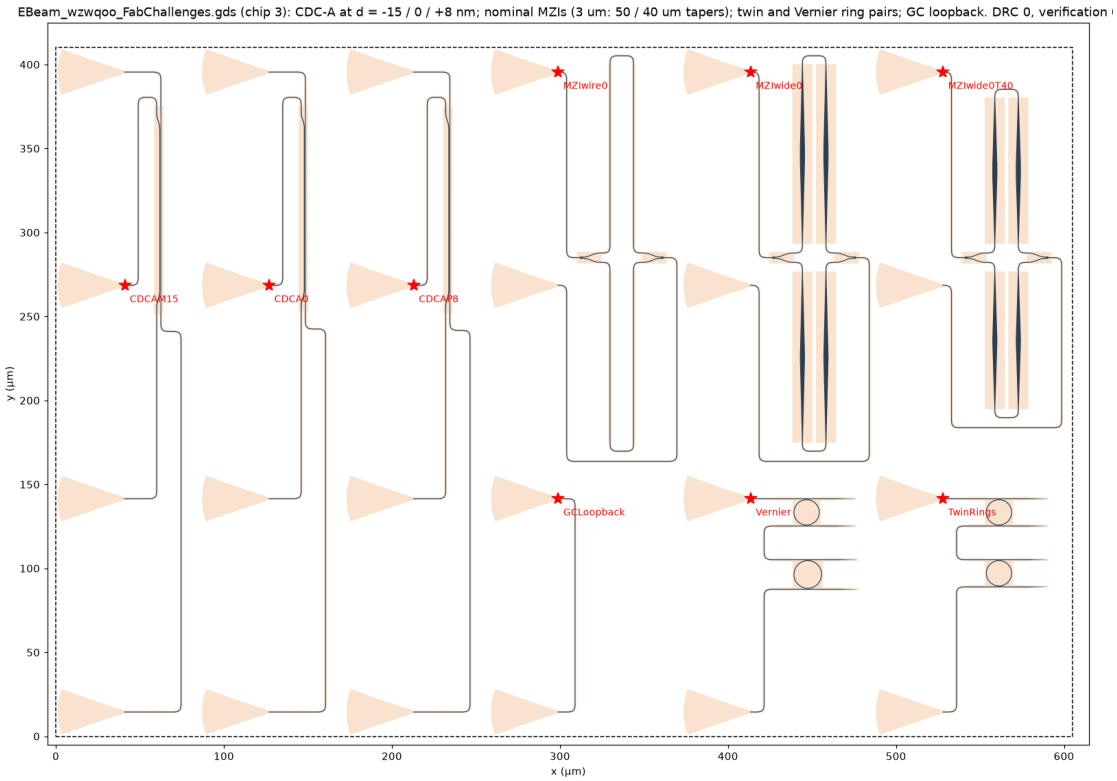


Fig. 7. Chip 3 (EBeam_wzwqoo_FabChallenges). CDC-A at −15 / 0 / +8 nm, each on four grating couplers; the nominal wire MZI above the GC loopback; the nominal 3 µm MZI with 50 µm and with 40 µm tapers, above the Vernier and twin ring pairs.

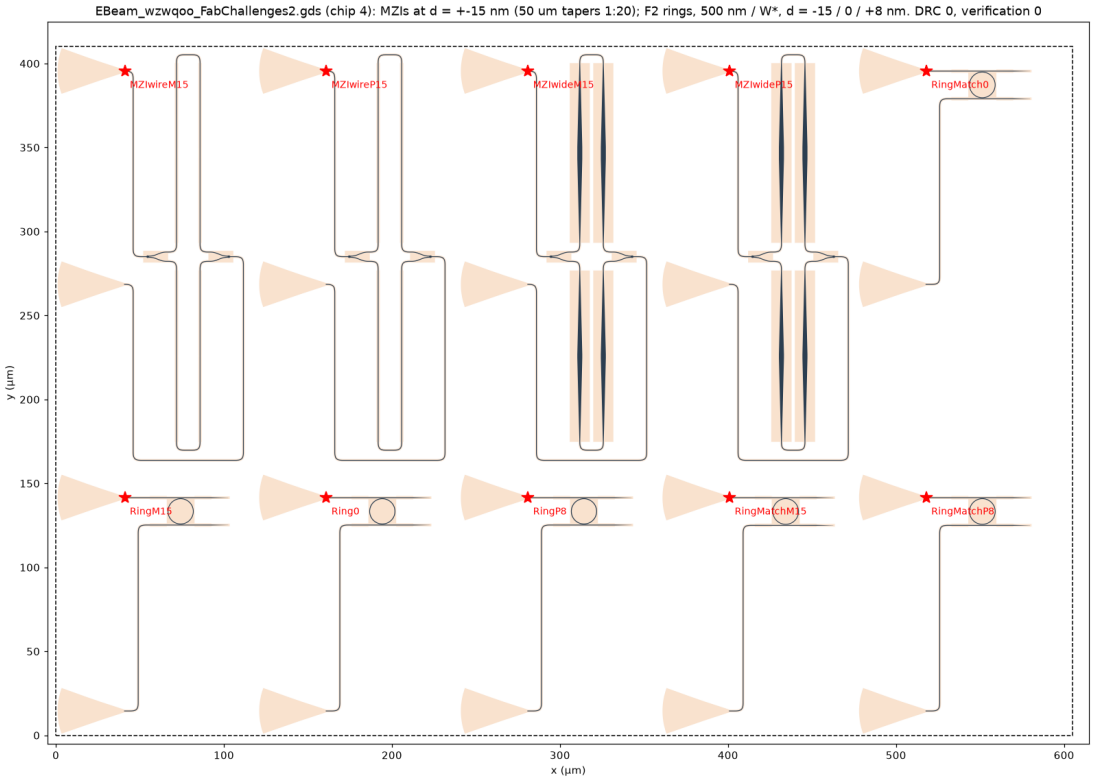


Fig. 8. Chip 4 (EBeam_wzwqoo_FabChallenges2). The wire and 3 µm MZIs at −15 and +15 nm, with the 500 nm ring at −15 / 0 / +8 nm and the W* ring at −15 nm under them; the W* ring at 0 and +8 nm in the right column.

are 4.9 / 7.3 / 3.2 / 5.7 µm apart on chips 1–4. The closest black boxes are 46.0 µm apart on chip 3, where the CDC columns sit 86 µm apart, and more than 60 µm elsewhere.

Fibre-array map. The fibre array has one input and three outputs at 127 µm pitch. Table 11 lists what each fibre carries; "—" is a fibre position with no coupler or with another device's coupler. Four devices measure only 3 of their 4 coarse channels, and the missing channel differs from device to device.

Table 11. Fibre-array map: fibre 1 (top) to fibre 4 (bottom) of each grating-coupler column

Chip	Device (label)	Fibre 1	Fibre 2	Fibre 3	Fibre 4	Not measured
1	LatticeD1	C3	input	C2	C4	C1 (terminated)
1	RingCDCD2	F3 (W4 drop)	input	F4 (CDC-B drop)	F1 + F2 (CDC-A drop)	—
1	RingCDCApo	F3 (W4 drop)	input	F4 (CDC-B drop)	F2 (CDC-A drop)	F1 (no W1)
2	AWGC	C4	input	C1	C2	C3 (terminated)
2	EDGC	C1	input	C4	C3	C2 (terminated)
2	CDCCascade	C4	input	C3	C2	C1 (no fourth CDC)
2	FPCavity	—	—	input	transmission	—
3	CDCAM15, CDCA0, CDCAP8	drop	input	through	add	—
3	MZIs (top pairs)	input	output	—	—	—
3	GCLoopback, ring pairs (bottom pairs)	—	—	input	output / drop	—
4	MZIs, RingMatch0 (top pairs)	input	output / drop	—	—	—
4	other rings (bottom pairs)	—	—	input	drop	—

The AWG-C and EDG-C channel order follows from the simulated output order and the layout: on the chip, AWG-C's outputs run C1, C2, C3, C4 from left to right and EDG-C's run C4, C3, C2, C1.

4.2. Verification

Table 12. Checks on the submitted files: the openEBL submission checks, SiEPIC functional verification and the EBeam DRC deck, run in KLayout on the files as submitted

File = top cell	Si extent (µm)	Black-box GCs	opt_in labels	DRC errors	Verification errors
EBeam_wzwqoo_design	555.1 × 392.0	12	3	0	0
EBeam_wzwqoo_benchmark	559.7 × 381.5	14	4	0	0
EBeam_wzwqoo_FabChallenges	558.2 × 391.2	24	9	0	0
EBeam_wzwqoo_FabChallenges2	539.1 × 391.2	20	10	0	0

- **SiEPIC-1a warnings.** The PDK's batch DRC deck adds one warning type, a pin marker not fully covered by silicon: 0 / 2 / 0 / 50 on chips 1–4. Chip 2's sit on the PDK Bragg cell's own pins at the FP cavity; chip 4's sit on the odd-nanometre widths (479, 485, 487, 515, 2985 and 3015 nm), drawn 1 nm narrower as described above.
- **Terminators face empty space.** ebeam_terminator_te1550 radiates the light forward from its tip, so every terminator that carries real power was checked for silicon within $\pm 20^\circ$ and 40 µm ahead of its tip. The AWG's and EDG's unused outputs, which carry a whole coarse channel, face the empty area beside the device.
- **Waveguide alignment.** Every component and waveguide pin meets exactly one partner at the same point (0 nm offset), facing the opposite way, with the same width: 132 / 42 / 142 / 168 pin pairs on chips 1–4, none misaligned, mismatched or unpaired. The cells counted in each layout equal the build plan, type by type.

5. Fabrication and Temperature Experiments

Experiment 1: does the F2 ring stay inside CDC-A's band? On chip 1 the F2 line sits only 0.48 nm inside CDC-A's upper 1 dB edge. By (1), a width error d moves both, but at different rates: 0.591 nm per nm of width for the ring and 0.667 nm/nm for the CDC. The margin is used up at $d \approx -6$ nm. A ring 479 nm wide (W*) moves at 0.662 nm/nm, nearly as fast as the CDC, and keeps F2 at every offset.

Experiment 2: a delay line that ignores width. In an MZI everything the two arms share (bends, splitters, equal straights) cancels. Only the extra length ΔL matters, so only the width of the guide that carries ΔL moves the fringes. Putting ΔL in a 3 µm guide instead of 500 nm wire makes the fringes 167 times less sensitive in simulation. ΔL = 10 µm

gives an FSR of about 57 nm, so each spectrum shows one or two nulls. The experiment tracks the position of one null, which moves by less than half an FSR over the $\pm$15 nm range, so the same null is identified on every copy.

Table 13. Chips 3 and 4 predictions (INTERCONNECT)

Device	d = −15 nm	d = 0	d = +8 or +15 nm
MZI null, delay in 500 nm wire	1527.06 nm	1536.03 nm	1544.20 nm (+15)
MZI null, delay in 3 µm guide	1568.09 nm	1568.14 nm	1568.19 nm (+15)
CDC-A 1 dB band	1526.23–1534.83 nm	1536.40–1545.00 nm	1541.58–1550.18 nm (+8)
F2 ring line, 500 nm	1535.44 nm (outside the band: F2 lost)	1544.52 nm	1549.02 nm (+8)
F2 ring line, 479 nm (W*)	1534.37 nm (0.46 nm inside)	1544.53 nm	1549.59 nm (+8)

Every column is on the die: the d = 0 copies are MZIwire0, MZIwide0, CDCA0, Ring0 and RingMatch0.

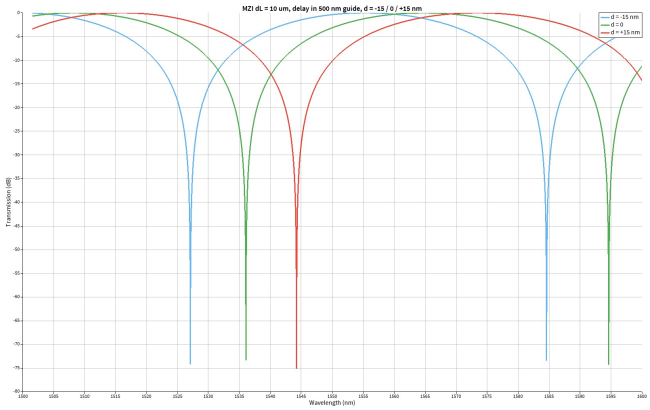


Fig. 9. MZI with its delay in 500 nm wire, d = −15 / 0 / +15 nm: the null moves about 9 nm per 15 nm of width.

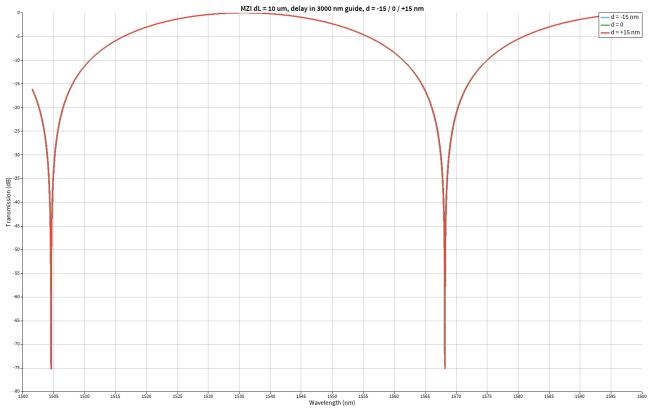


Fig. 10. The same MZI with its delay in a 3 µm guide: the three curves lie within 0.10 nm.

How chips 3 and 4 are read. The copies of each device share the same unknown fabrication bias, so the slope between the two offset copies (wavelength shift divided by the 23 or 30 nm width difference) does not depend on that bias. The d = 0 copy adds a third point on every device type.

Limits of the width-offset reading.

- **The responses are not linear.** The simulated responses bend slightly. Table 14 compares the slope on each side of d = 0 with the slope between the two offset copies: they differ by up to $\pm$**5 %**. With the d = 0 copy on the chip, this curvature is measured as well.
- **Shared bias.** All copies sit on the same die, so an unknown fabrication bias b shifts every width alike. It cancels in the difference, but the slope is then the slope at b instead of at 0. With the curvature of Table 14, a bias of **10 nm** changes the measured slope by **4–6 %**, and the ring-to-CDC slope difference (0.077 nm/nm) by $\pm$**0.011 nm/nm**. The wire-to-wide ratio of the MZIs (about 170$\times$) is far larger than these errors.
- **Variation within the die.** Neighbouring devices may differ for reasons other than d: columns sit 86–120 µm apart, and local dose, resist and film thickness vary. The wide-MZI pair is the most exposed: its predicted 0.10 nm difference equals what a 0.1 nm thickness difference between the two devices would produce (1.37 nm per nm of thickness). The ring pairs of Experiment 3 measure this variation directly.

Experiment 3: two ring pairs. Two add-drop rings in series pass light only where a resonance of each falls. By (5) ring 1 (R 7.529 µm) repeats every 12.06 nm and Vernier's ring 2 (R 8.232 µm) every 11.03 nm, so their resonances coincide every

$$\mathrm{FSR}_V = \frac{\mathrm{FSR}_1\,\mathrm{FSR}_2}{\mathrm{FSR}_1 - \mathrm{FSR}_2} = \frac{12.06 \times 11.03}{1.03} \approx 129 \text{ nm} \tag{14}$$

which is wider than the 102 nm sweep, so Vernier shows one line. If ring 2's comb shifts by δ against ring 1's, the coincidence moves by

Table 14. Slope of each width-test feature on either side of d = 0 (from Table 13), and how much a shared 10 nm bias moves the slope between the offset copies

Feature	Slope −15 → 0 nm	Slope 0 → +8 / +15 nm	Slope between offset copies	Change per 10 nm of bias
MZI null, delay in 500 nm wire	0.598 nm/nm	0.545 nm/nm (+15)	0.571 nm/nm	0.036 nm/nm (6 %)
MZI null, delay in 3 µm guide	0.0033 nm/nm	0.0033 nm/nm (+15)	0.0033 nm/nm	< 0.001 nm/nm
F2 ring line, 500 nm	0.605 nm/nm	0.563 nm/nm (+8)	0.590 nm/nm	0.037 nm/nm (6 %)
F2 ring line, 479 nm (W*)	0.677 nm/nm	0.633 nm/nm (+8)	0.662 nm/nm	0.039 nm/nm (6 %)
CDC-A upper 1 dB edge	0.678 nm/nm	0.648 nm/nm (+8)	0.667 nm/nm	0.027 nm/nm (4 %)

$$\Delta\lambda_V = \frac{\mathrm{FSR}_1}{\mathrm{FSR}_1 - \mathrm{FSR}_2}\,\delta \approx 11.7\,\delta \tag{15}$$

- **TwinRings, the size of a local difference.** Its rings are identical, so by (6) a shift δ dims every line by $1/[1 + (2\delta/\mathrm{FWHM})^2]$, with FWHM = 0.19 nm. δ = 0.1 nm, a local width difference of 0.17 nm, dims every line by 3.3 dB; a 0.5 dB dimming corresponds to $\delta \approx 0.03$ nm. All eight orders in the sweep give the same reading, and Ring0 and the simulated height (−0.56 dB) are the references.
- **Vernier, its sign.** The same δ moves the coincidence 11.7 times as far, towards red or blue depending on which ring printed wider. The transmitted lines always sit on ring 1's resonances, so the heights of the brightest line and its two neighbours place the coincidence. In simulation it passes one line at **1544.53 nm**, 0.12 nm wide, at −0.59 dB; ring 1's neighbouring orders miss ring 2's by 1.03 nm and pass at −**20.7 dB**.
- **Why not two rings of different width?** Two rings of equal radius whose widths differ by 2 nm have nearly equal FSRs (n_g differs by 0.08 %); their combs stay about 1.2 nm apart across the whole sweep and never coincide. A single line needs two different FSRs, which only the radius provides.

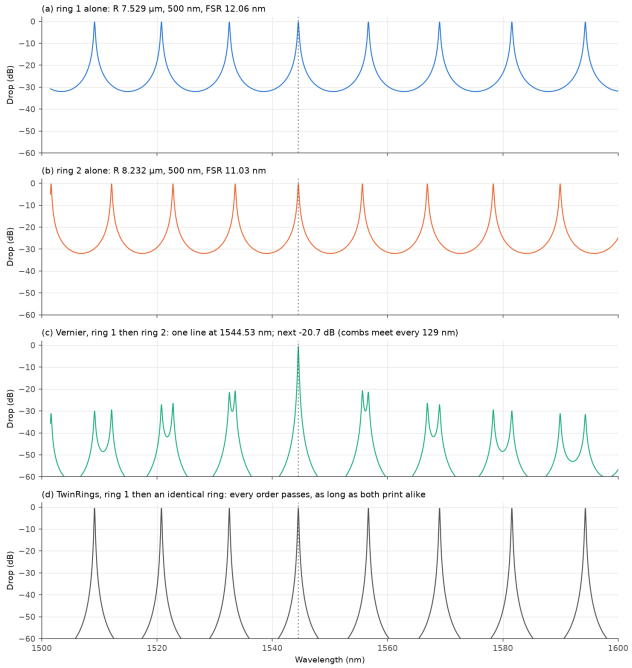


Fig. 11. Chip 3's ring pairs in INTERCONNECT. (a) Ring 1 alone, R 7.529 µm: a comb 12.1 nm apart. (b) Ring 2 alone, R 8.232 µm: a comb 11.0 nm apart. (c) Vernier, ring 1 then ring 2: the combs meet only at F2 (dotted), one line in the sweep. (d) TwinRings, ring 1 then an identical ring: every order passes.

Experiment 4: temperature. openEBL can measure devices a second time, about 10 K warmer. Six devices are requested: RingCDCD2, FPCavity, and on chip 3 CDCA0, GCLoopback, Vernier and MZIwide0. By (1), with dn_{eff}/dT = 1.9

$\times$ 10^{-4} K^{-1} for every guide, each feature should move as in Table 15. The measured shift gives the thermo-optic coefficient of the fabricated guide:

$$\frac{dn_{\text{eff}}}{dT} = \frac{n_g}{\lambda}\frac{\Delta\lambda}{\Delta T} \tag{16}$$

Table 15. Predicted shifts for ΔT = 10 K

Device	Feature	Room temperature	Drift rate	At +10 K
RingCDCD2	ring lines F1–F4	1542.94–1547.72 nm	0.0701–0.0704 nm/K	each line +0.70 nm
CDCA0	1 dB band, peak	1536.4–1545.0 nm, peak 1542.8 nm	0.0707 nm/K	1537.1–1545.7 nm, peak 1543.5 nm
FPCavity	transmission line	1544.28 nm	0.0702 nm/K	1544.98 nm
Vernier	the single line	1544.53 nm	0.0702 nm/K	1545.23 nm, same ring-1 order
MZIwide0	MZI null	1568.14 nm	0.0803 nm/K	1568.94 nm
GCLoopback	GC envelope peak	1554.5 nm, -4.71 dB	not predicted	measured; normalises every warm spectrum

- **Rings and CDC move together.** The ring and CDC drift rates differ by 0.0005 nm/K, 0.005 nm over 10 K, about 1 % of F2's margin. The width mismatch of Experiment 1 is 0.077 nm per nm, so the whole 10 K swing misaligns them as much as a 0.065 nm width error would: temperature is the benign variation, width the one that sets the design rule.
- **Three checks.** GCLoopback measures the grating couplers' own shift. Vernier's line should move with F2 and keep its ring-1 order; a jump to the next order would mean the two rings' dn_{eff}/dT differ. MZIwide0 ignores width but not temperature: its null should move 0.80 nm, as far as any guide's.

6. Test Results

Measurements pending: this section will hold the openEBL results.

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